

request&json

May 10, 2026

```
[ ]: ## requests and json
import requests
import json
import pandas as pd
# Basic GET request
url = "https://jsonplaceholder.typicode.com/posts/1"
response = requests.get(url)
print(response)
# Check status code
print(response.status_code)
# View raw response text
print(response.text)
# Convert response to JSON
data = response.json()
print(data)
print(type(data))
# Access JSON values
print(data["userId"])
print(data["title"])
print(data["body"])
# Pretty print JSON
formatted_json = json.dumps(data, indent=4)
print(formatted_json)
# Get multiple posts
url = "https://jsonplaceholder.typicode.com/posts"
response = requests.get(url)
posts = response.json()
print(type(posts))
print(len(posts))
# Loop through JSON list
for post in posts[:5]:
    print(post["id"])
    print(post["title"])
    print("-" * 50)
# Send query parameters
params = {
    "userId": 1
```

```

}
response = requests.get(
    "https://jsonplaceholder.typicode.com/posts",
    params=params
)
print(response.url)
filtered_posts = response.json()
print(len(filtered_posts))
# Send custom headers
headers = {
    "User-Agent": "Mozilla/5.0",
    "Accept": "application/json"
}
response = requests.get(
    "https://jsonplaceholder.typicode.com/posts/1",
    headers=headers
)
print(response.status_code)
# Send POST request
new_post = {
    "title": "Learning Requests",
    "body": "This is tutorial content",
    "userId": 1
}
response = requests.post(
    "https://jsonplaceholder.typicode.com/posts",
    json=new_post
)
print(response.status_code)
print(response.json())
# Send PUT request
updated_post = {
    "id": 1,
    "title": "Updated Title",
    "body": "Updated body",
    "userId": 1
}
response = requests.put(
    "https://jsonplaceholder.typicode.com/posts/1",
    json=updated_post
)
print(response.json())
# Send PATCH request
patch_data = {
    "title": "Patched Title"
}
response = requests.patch(

```

```

        "https://jsonplaceholder.typicode.com/posts/1",
        json=patch_data
    )
    print(response.json())
    # Send DELETE request
    response = requests.delete(
        "https://jsonplaceholder.typicode.com/posts/1"
    )
    print(response.status_code)
    # View response headers
    response = requests.get(
        "https://jsonplaceholder.typicode.com/posts/1"
    )
    print(response.headers)
    # Check content type
    print(response.headers["Content-Type"])
    # Download image using binary content
    image_url = "https://picsum.photos/200"
    response = requests.get(image_url)
    with open("random_image.jpg", "wb") as f:
        f.write(response.content)
    print("Image downloaded")
    # Save JSON data to file
    response = requests.get(
        "https://jsonplaceholder.typicode.com/posts/1"
    )
    data = response.json()
    with open("post.json", "w") as f:
        json.dump(data, f, indent=4)
    print("JSON file saved")
    # Read JSON data from file
    with open("post.json", "r") as f:
        loaded_data = json.load(f)
    print(loaded_data)
    # Convert API response to DataFrame
    response = requests.get(
        "https://jsonplaceholder.typicode.com/posts"
    )
    posts = response.json()
    df = pd.DataFrame(posts)
    print(df.head())
    # Loop through API data
    for post in posts[:5]:
        print("ID:", post["id"])
        print("TITLE:", post["title"])
        print("BODY:", post["body"])
        print("=" * 50)

```

```

# Get nested JSON response
response = requests.get(
    "https://jsonplaceholder.typicode.com/users/1"
)
user = response.json()
print(user)
# Access nested JSON values
print(user["name"])
print(user["email"])
print(user["address"]["city"])
print(user["company"]["name"])
# Convert JSON string into Python object
json_string = '''
{
    "name": "Alice",
    "age": 25
}
'''
python_data = json.loads(json_string)
print(python_data)
print(type(python_data))
# Convert Python object into JSON string
person = {
    "name": "Bob",
    "age": 30
}
json_output = json.dumps(person, indent=4)
print(json_output)
# Show common HTTP status codes
print("200 = OK")
print("201 = Created")
print("400 = Bad Request")
print("401 = Unauthorized")
print("403 = Forbidden")
print("404 = Not Found")
print("500 = Server Error")

```